

CENG322 - Video Assignment 2

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1. What are the advantages of heterogeneous/asymmetric computing? Give examples.

Heterogeneous/Asymmetric computing is using different types of processors within the same system to better handle different types of tasks. Not every workload needs the same type of processing resources. Some tasks are best done with high-performance cores, while others work well on simpler, more efficient cores.

- Different parts of an application may have different needs. For example, a large core may process serial parts of a program faster, while a small core may process parallel parts more efficiently. This results in better performance and energy savings.
- Another advantage is flexibility: systems can adapt to different workloads by scheduling tasks to the most suitable cores.

A real-world example would be modern mobile processors, such as Apple's M-series or ARM-based systems, which contain both performance and efficiency cores.

2. What are the overheads causing imperfect parallelism in parallel part of a program? Explain each.

Even if a program can be parallelized, there are several types of overhead that can prevent scaling:

- Synchronization: When multiple threads need to coordinate, one thread may have to wait for the others, which reduce the parallel efficiency.
- Serialization: Some parts of a program cannot be parallelized. According to Amdahl's Law, these parts limit the speedup no matter how many cores are added.
- Load imbalance: Occurs when work is not distributed equally among threads. Some threads may finish early while others are still running. This imbalance reduces performance.
- Resource contention: Threads may compete for shared hardware resources, such as caches and memory bandwidth. This can cause additional waits and delays.

3. What are the topics already given in our parallelism lecture material ? What are the topics completely new to you in the video?

The topics we have seen in class:

- Multithreading and thread creation (Pthreads, Windows threads)
- Multicore programming: task parallelism and data parallelism
- Amdahl's Law
- Synchronization, load balancing and resource sharing

The topics new to me: Bottlenecks; cooperation between hardware and software to detect bottlenecks during runtime.

4. After watching the video, which specific paper (mentioned in the video) do you want to read the most? Why?

I wanted to read the "*Bottleneck Identification and Planning*" article. It would be informative since I don't know much about it. The video explains how it identifies bottlenecks that cause the system to wait. It then uses powerful cores to accelerate them.